

Nate Cirino

832-833-5202 | cirino.na@northeastern.edu | [portfolio](#) | [linkedin.com/in/nate-cirino](https://www.linkedin.com/in/nate-cirino) | github.com/nateci

Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Digital Design & Computer Architecture, Systems Security, Cybersecurity, Computer Networks, Algorithms & Data Structures, Foundations of Data Science, Object-Oriented Design, Foundations of Software Engineering

PROJECTS

Kepler | *Rust, C++, LSM-tree storage engine, Raft, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust around a from-scratch LSM-tree storage engine
- Implemented the on-disk write path end to end: memtable, write-ahead log, and leveled SSTable persistence
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution
- Implemented Raft election, log replication, and membership changes plus an MVCC snapshot-isolation layer

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Preprocessed thermal sensor spectrograms and trained/evaluated CNN models detecting novel aquatic signatures
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Applied AI agents to engineering workflows: autonomous incident triage via tool-use over Prometheus and Loki
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair physical retail brands: e-labels, mobile, and in-store POS
- Building a supplier-agnostic e-label device API unifying 4 hardware suppliers behind one interface
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Applied secure coding practices implementing OIDC/IdP auth via Azure across 20 health systems, 10K+ clinicians
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Improved development tooling and process: CI/CD workflows on Azure DevOps and K8s with automated rollback
- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

SKILLS & HONORS

Languages/Tools: C, C++, Rust, Python, Bash, Linux, Java, TypeScript, JavaScript, Docker, Kubernetes, Jenkins, Buildkite, Terraform, Groovy, AWS, Azure, PostgreSQL, GraphQL

AI/ML & Data: PyTorch, TensorFlow, Pandas, SageMaker, AWS Neuron SDK, pgvector, Bedrock KB

Frameworks & Libraries: Flask, FastAPI, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Tech Lead at Northeastern Electronic Racing, Dean's List (All Semesters), PawHacks, json-io OSS Contributor